



2025

I22

BOOKLET NO.

136057

Chemistry
Paper – I

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.



SECTION – A

- Q1.** Solve the following. **10×5=50**
- Q1.** (a) Define ideal and real gas. Explain how real gases deviate from ideal behaviour. What are the causes of deviation ? **10**
- Q1.** (b) What are the effects of surface tension in daily life ? Calculate the surface tension of toluene in which it rises 2 cm in capillary tube of radius 0.035 cm. The density of toluene at 20°C is 0.860 gm/cc ($g = 9.8 \text{ ms}^{-2}$). **10**
- Q1.** (c) State first Law of Thermodynamics in three different forms. Write down its mathematical equation with their usual meanings. Critically examine
- a) isothermal
 - b) adiabatic and
 - c) cyclic processes. **10**
- Q1.** (d) Explain construction, working, advantages and disadvantages of glass electrode. **10**
- Q1.** (e) Discuss Transition state theory. Give its advantages over the Collision theory. **10**
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- Q2.** (a) Define entropy. Explain how entropy changes take place in a reversible process and in an irreversible process. Comment on entropy of the universe is increasing and calculate entropy change in one mole of an ideal gas expands from a volume of 5 dm^3 to a volume of 50 dm^3 in an isothermal reversible process at 300 K. (Given : $R = 8.314 \text{ JK}^{-1}.\text{mol}^{-1}$). **20**
- Q2.** (b) What are the amperometric titrations ? Explain the nature of curves obtained in amperometric titration. Give in short its advantages, disadvantages and applications and also explain when the dead stop end point method is applicable. **20**
- Q2.** (c) Draw a labelled phase diagram of Water system and discuss its salient features. **10**
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- Q3.** (a) What is Joule-Thomson effect ? Justify that during this process, enthalpy of the system remains constant. Also explain what you understand by Joule-Thomson coefficient and Inversion temperature. Derive that Joule-Thomson coefficient is given by $\mu = -\frac{1}{C_p} \left(\frac{\partial H}{\partial P} \right)_T$. **20**
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- Q3.** (b) i) Define second order reaction and derive the expression for the rate constant of second order reaction $2A \rightarrow \text{product}$.
ii) A second order reaction in which the initial concentration of both the reactants are same is 25% complete in 600 seconds. How long will it take for the reaction to go to 50% completion ? **10**
- Q3.** (c) Explain decay of excited states by different routes using Jablonski diagram. **10**
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- Q3.** (c) State and explain Laws of Photochemistry. **10**
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- Q4.** (a) Explain relaxation and electrophoretic effect as used in Debye-Huckel theory. Write the Debye-Huckel-Onsager equation with their different meaning of different quantities. The viscosity of water at 0°C and at 100°C are 0.01793 and 0.00284 poise respectively. The corresponding dielectric constants are 87.8 and 56. Calculate the Onsager constant A and B for 1 : 1 electrolyte. **20**
- Q4.** (b) Define quantum yield. What are the reasons for the low and high quantum yields ?
A certain system absorbs 2×10^{16} quantum of light per second. On irradiation for 10 minutes 0.001 mole of the reactant was found to have reacted. Calculate the quantum yield for the process.
(Given, Avogadro's no. = 6.023×10^{23}) **20**
- Q4.** (c) Define adsorption isotherm. What are the assumptions of Langmuir and deduce the Langmuir adsorption isotherm. **10**
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SECTION – B

- Q5.** Solve the following. **10x5=50**
- Q5.** (a) State and explain Heisenberg's uncertainty principle. Find the uncertainty in position of an electron having an uncertainty in velocity is $6 \times 10^{-6} \text{ m/sec}$.
(Given $m_e = 9.1 \times 10^{-31} \text{ Kg}$, $h = 6.626 \times 10^{-34} \text{ Kg.m}^2.\text{sec}^{-1}$) **10**
- Q5.** (b) Draw the molecular orbital diagram for He_2^+ and HF and calculate their bond order. **10**
- Q5.** (c) Explain body centered cubic (bcc) and face-centered cubic (fcc) lattice with respect to cesium chloride and rock salt. **10**
- Q5.** (d) Describe the role of metal ions in biological systems focusing on their function in ion transport across membranes with suitable examples. **10**
- Q5.** (e) What is trans effect ? Give its application towards the synthesis of cis and trans-platin. **10**
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- Q6. (a) i)** What is Born-Haber cycle ? Discuss its usefulness by explaining the stability of solids.
- ii)** Calculate the lattice energy of sodium chloride (in kJ/mol) from the following data.
- $A = 1.75$ $r_0 = 2.8 \text{ \AA}$
 $n = 9$ $N = 6.023 \times 10^{23}$
 $e = 4.8 \times 10^{10} \text{ e.s.u.}$
 $1\text{J} = 10^7 \text{ ergs}$ **20**
- Q6. (b)** What are the assumptions of Valence Bond Theory (VBT) for co-ordination compounds ? Draw the VB diagram and indicate the type of hybridization, shape and magnetic character for each of the following.
- a) $[\text{Ni}(\text{NH}_3)_4]^{2+}$
b) $[\text{Pt}(\text{NH}_3)_4]^{2+}$ and
c) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ **20**
- Q6. (c)** Why Noble gases are unreactive and explain why only Xenon and Krypton form compounds ? Give the structures and properties of XeF_4 . **10**
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- Q7. (a)** What do you mean by EAN rule and metal carbonyls ? How are metal carbonyls prepared ? Explain nature of bonding in metal carbonyls with the help of synergic mechanism. **20**
- Q7. (b)** What are interhalogen compounds ? Explain two methods of preparation of AX and AX_3 type with their properties, uses and structures. Why are interhalogens more reactive than halogens ? **20**
- Q7. (c)** What is Lanthanide contraction ? What are the effects of Lanthanide contraction ? **10**
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- Q8. (a)** Derive Bragg's equation for X-ray diffraction. What is the maximum wavelength permissible for first order diffraction in (100) plane of a simple cubic lattice of side length 'a' ? The first order reflections from the 100, 110 and 111 planes of a given cubic crystal (NaCl) were found to occur at angles 5.9° , 8.4° and 5.2° respectively. Determine the type of cubic lattice to which the crystal belongs. (Given $\sin 5.9^\circ = 0.103$, $\sin 8.4^\circ = 0.146$, $\sin 5.2^\circ = 0.094$) **20**
- Q8. (b)** What are silicones ? What is general formula of it ? Explain their preparation and discuss different types of silicones with examples. **20**
- Q8. (c)** What are the assumptions of Crystal field theory ? Explain Crystal field splitting of d orbitals in octahedral complex with suitable examples. **10**
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